

JDBC & Database Connectivity

Assignment: JDBC-Based Student CRUD Application

Objective:

To design and implement a Java application using JDBC (Java Database Connectivity) that performs **CRUD (Create, Read, Update, Delete)** operations on a **MySQL database**. This task reinforces JDBC concepts, database connection handling, SQL interaction, and clean code structure using PreparedStatements.

Problem Statement:

You are required to build a console-based **Student Management System** using Java and MySQL. The application should support the following operations on a **Student** table in the **SchoolDB** database:

1. **Insert (Create)** a new student record
2. **Read** (fetch) a student record by ID
3. **Update** a student's name by their ID
4. **Delete** a student record by ID
5. **Exit** the application

Use **JDBC** for all interactions with the database, and ensure the use of **PreparedStatement** to handle dynamic inputs and prevent SQL injection.

Database Schema

Assume the following structure for your **Student** table in the **SchoolDB** database:

```
CREATE DATABASE SchoolDB;
```

```
USE SchoolDB;
```

```
CREATE TABLE Student (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    age INT,  
    grade VARCHAR(10),  
    email VARCHAR(100)  
);
```

Functional Requirements:

1. Create Student

- Prompt user to enter **name**, **age**, **grade**, and **email**
- Insert a new row into the **Student** table

2. Read Student

- Prompt user to enter the student **id**
- Display all details of the student if found

3. Update Student Name

- Prompt for **id** and the new **name**
- Update the student's name where ID matches

4. Delete Student

- Prompt for **id**
- Delete the student record with the specified ID

5. Exit

- Terminate the program
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Technical Requirements:

- Use **JDBC** to connect to the database
- Use **PreparedStatement** for all SQL queries
- Handle exceptions using try-catch blocks or try-with-resources
- Use a **menu-driven** structure (infinite loop with switch-case)
- Ensure that all resources (**Connection**, **PreparedStatement**, **ResultSet**) are closed properly.